BANK CUSTOMER CHURN ANALYSIS USING PYTHON

Exploratory and Statistical Investigation of Customer Attrition Patterns, Banking Behaviour, and Retention Opportunities

Revised Project Documentation

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

1. Project Overview

This project analyzes bank customer data to explore customer churn patterns and identify characteristics associated with customers leaving the bank. Using Python, the analysis examines customer demographics, geographical differences, account balances, and membership activity to understand how churn varies across customer groups.

The project applies data inspection, data cleaning and transformation, statistical analysis, and data visualization to convert raw customer records into meaningful findings and business recommendations. The completed work is exploratory and statistical; it does not include a machine-learning churn prediction model.

2. Project Objective

To analyze bank customer churn using Python by exploring customer demographics, credit scores, account balances, estimated salaries, and banking behaviour. The project aims to identify customer attrition patterns, examine characteristics associated with churn, and generate evidence-based insights that can support customer-retention decisions.

3. Problem Statement

Customer churn is an important concern in banking because losing customers can affect long-term customer relationships and business performance. This project investigates which customer characteristics and banking behaviours are associated with higher customer churn, and what retention opportunities can be identified from the available data.

The dataset records whether a customer stayed or exited, but it does not contain the customer's stated reason for leaving. Therefore, the analysis identifies associations and areas for further investigation rather than proving causes of churn.

4. Dataset Information

- Dataset: Bank Customer Churn
- Source: Maven Analytics Data Playground
- Domain: Banking and Customer Analytics
- File format: CSV
- Dataset size: 10,000 customer records and 13 original columns
- Analysis environment: Google Colab
- Programming language: Python
- Target variable: Exited (0 = Stayed; 1 = Exited)
- Data collection period: Not specified in the supplied dataset information

5. Dataset Attributes

Attribute	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer credit score
Geography	Customer's country
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
NumOfProducts	Number of banking products held
HasCrCard	Credit card ownership indicator
IsActiveMember	Active membership indicator
EstimatedSalary	Estimated customer salary
Exited	Customer churn indicator

6. Tools and Technologies

- Python – programming and analysis
- Pandas – data loading, cleaning, grouping and summaries
- NumPy – numerical operations
- Matplotlib – chart creation
- Seaborn – statistical visualizations
- Google Colab – notebook development and execution
- GitHub – source-code storage and version control

7. Project Workflow

Stage 1: Initial Exploratory Data Analysis

The dataset was loaded from GitHub into a Pandas DataFrame. Initial exploration included previewing records, checking dataset dimensions, column names and data types, generating descriptive statistics, and checking missing values, duplicate records and the target-variable distribution.

Stage 2: Data Cleaning, Transformation and Feature Engineering

The dataset was checked for duplicates and missing values. The final notebook reports 10,000 rows, 15 columns, zero duplicate rows and zero missing values after preparation. Column names were standardized, and the derived features churn_status and age_group were created. Data types were inspected. Potential age outliers were identified and retained because they may represent valid customers.

Stage 3: Statistical Analysis and Visualization

Statistical analysis included measures of central tendency, variance, standard deviation, skewness, kurtosis and age outlier detection using a box plot and the IQR method. Nine objective-based visualizations were created: three univariate, four bivariate and two multivariate charts. Each visualization below includes the objective, chart, interpretation, key insight and business insight as requested in the project review guidance.

8. Statistical Analysis

8.1 Measure of Central Tendency

Objective: To calculate mean, median and mode for credit score, age, account balance and estimated salary to understand the typical characteristics of the customer base.

Variable	Mean	Median	Mode
Credit Score	650.53	652.00	850
Age	38.92	37.00	37
Account Balance	76,485.89	97,198.54	0.00
Estimated Salary	100,090.24	100,193.91	24,924.92

Interpretation: The average customer age is 38.92 years, with a median and mode of 37 years. Mean and median credit scores are close (650.53 and 652), as are mean and median estimated salaries (100,090.24 and 100,193.91). The mean account balance (76,485.89) is below the median (97,198.54), while the mode is zero, indicating that zero balances are the most frequently occurring balance value.

Key Insight: The dataset's typical customer is in their late 30s. Account balances show a difference between the mean and median, suggesting variation in customer balances.

Business Insight: These statistics provide a baseline customer profile. Comparing these variables by churn status helps identify whether retained and exited customers have different profiles.

8.2 Variance and Standard Deviation

Objective: To measure the spread of customer age, credit score, account balance and estimated salary around their respective means.

Variable	Variance	Standard Deviation
Credit Score	9,341.86	96.65
Age	109.99	10.49
Account Balance	3,893,436,175.99	62,397.41
Estimated Salary	3,307,456,784.13	57,510.49

Interpretation: Customer ages typically vary by 10.49 years around the mean age, while credit scores have a standard deviation of 96.65 points. Account balances and estimated salaries have standard deviations of 62,397.41 and 57,510.49, respectively, showing considerable spread in their original units.

Key Insight: Customer financial characteristics vary across the dataset. Standard deviations should be interpreted within each variable's own units and not compared as if all variables used the same scale.

Business Insight: Understanding the spread of customer financial profiles can help the bank consider customer segments separately rather than relying only on overall averages.

8.3 Skewness, Kurtosis and Outlier Detection

Objective: To examine distribution shape and identify potential outliers in numerical customer attributes.

Variable	Skewness	Excess Kurtosis
Credit Score	-0.072	-0.426
Age	1.011	1.395
Account Balance	-0.141	-1.489
Estimated Salary	0.002	-1.182

Interpretation: Age has positive skewness (1.011), indicating a longer right tail toward older customers. Credit score (-0.072) and account balance (-0.141) are close to symmetric with slight negative skew. Estimated salary (0.002) is almost symmetric. The negative excess kurtosis values for credit score, balance and salary indicate flatter distributions with lighter tails than a normal distribution; age's positive excess kurtosis indicates heavier tails and a more pronounced peak.

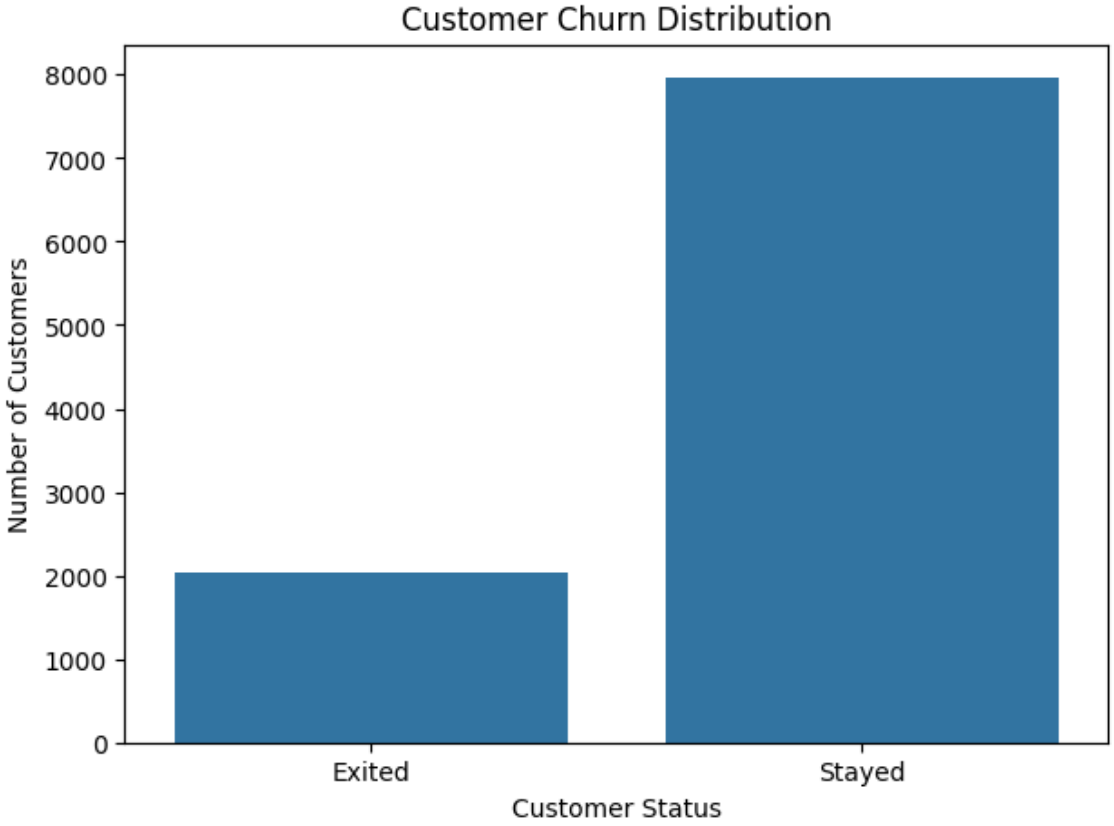
Key Insight: Age is the most positively skewed of the four variables and has positive excess kurtosis. The IQR analysis of age produced Q1 = 32, Q3 = 44 and IQR = 12, with an upper bound of 62 years. It identified 359 potential outliers above 62 years.

Business Insight: Retain these ages unless they are confirmed data errors. The older-customer segment can be examined separately to determine whether its churn rate differs from other age groups.

9. Objective-Based Visualizations and Interpretations

Univariate Analysis

Visual 1 — Customer Churn Distribution



Objective: To measure the overall distribution of customers who stayed with the bank and customers who exited.

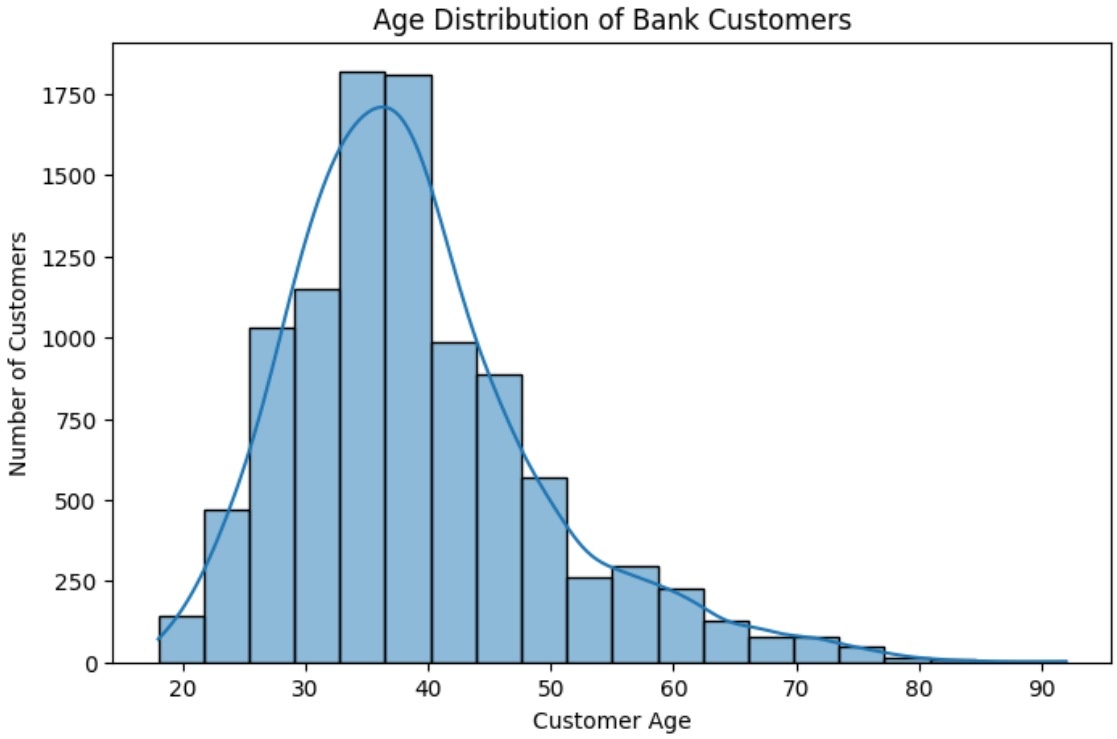
Visualization: Visual 1 — Customer Churn Distribution (chart screenshot from the Python notebook).

Interpretation: Out of 10,000 customers, 7,963 (79.63%) stayed and 2,037 (20.37%) exited. The exited group represents approximately one in five customers in this dataset.

Key Insight: The overall customer churn rate is 20.37%, while the retention rate is 79.63%.

Business Insight: Use 20.37% as a baseline for monitoring churn and comparing customer segments. The bank can investigate which segments contribute most to the exits.

Visual 2 — Age Distribution of Customers



Objective: To examine how customer ages are distributed and identify the age ranges in which customers are concentrated.

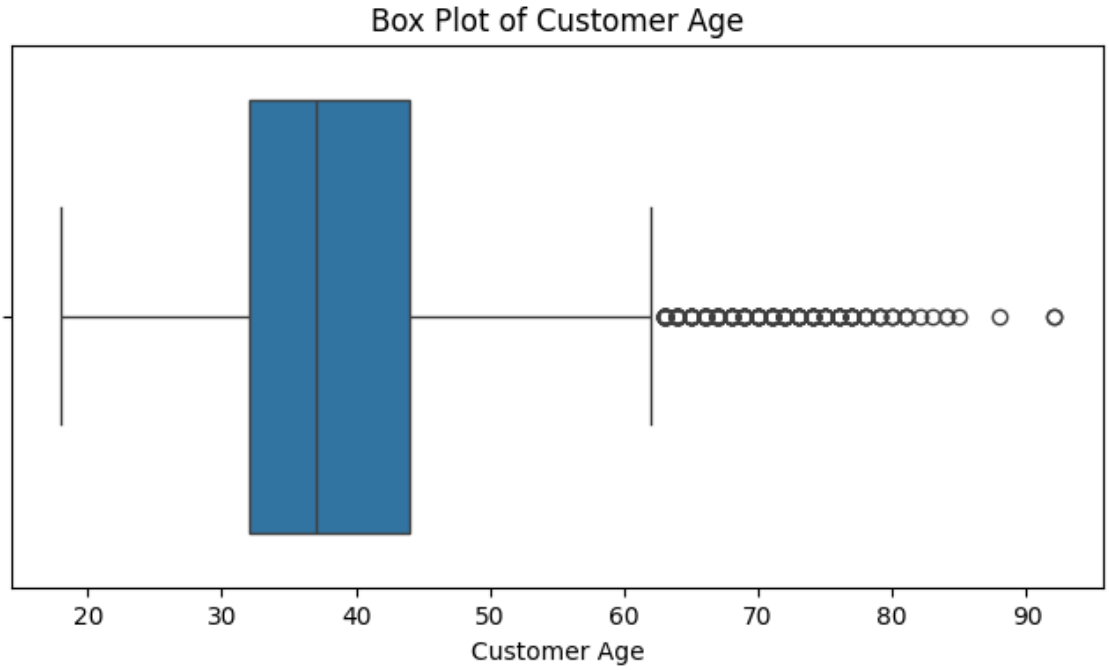
Visualization: Visual 2 — Age Distribution of Customers (chart screenshot from the Python notebook).

Interpretation: The mean customer age is 38.92 years and the median is 37 years. The histogram shows the concentration of customers around younger and middle-adult ages, with fewer observations at older ages. The positive skewness of 1.011 indicates a longer tail toward older customers.

Key Insight: The typical customer is in their late 30s, and the age distribution is positively skewed.

Business Insight: The bank can use age segments to compare product usage, service needs and churn rates. Age alone should not be treated as a reason for leaving.

Visual 3 — Box Plot of Customer Age



Objective: To summarize the median and spread of customer ages and identify potential age outliers.

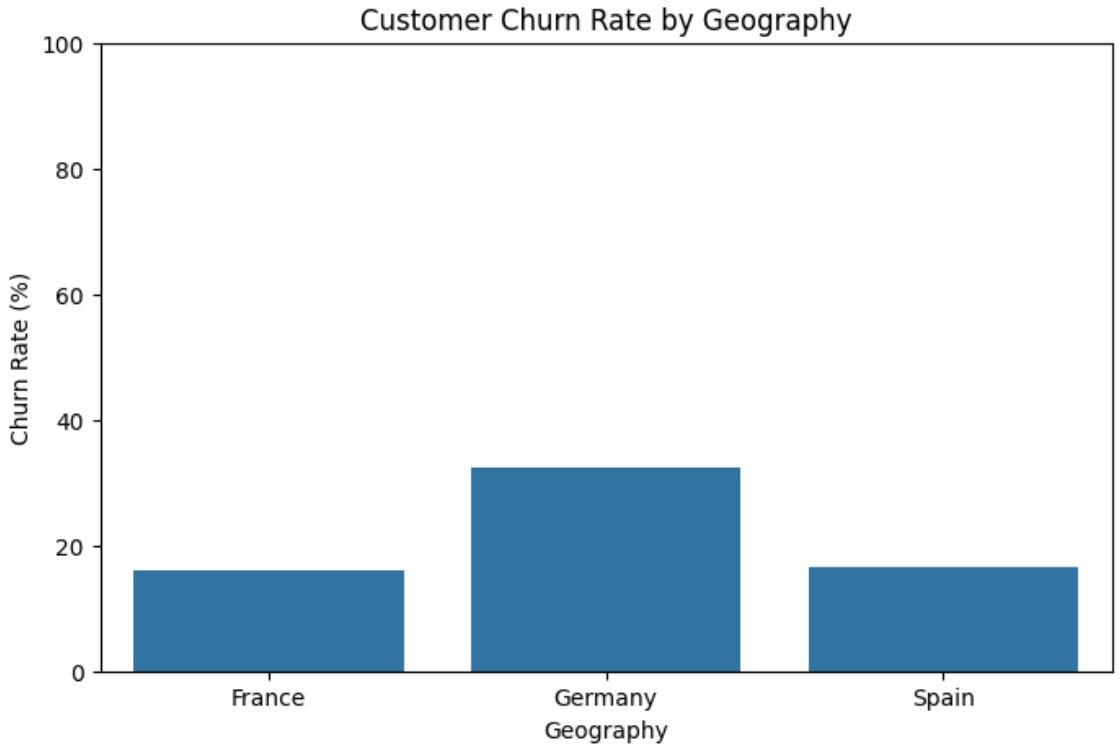
Visualization: Visual 3 — Box Plot of Customer Age (chart screenshot from the Python notebook).

Interpretation: The median customer age is 37 years, and the middle 50% of customers are between 32 and 44 years. The IQR is 12 years, giving an upper outlier boundary of 62 years. The IQR method identified 359 customers above 62 years as potential outliers.

Key Insight: Most customers are concentrated between 32 and 44 years, while 359 records are above the IQR upper bound of 62 years.

Business Insight: Retain these observations unless they are confirmed data errors. The older-customer segment can be examined separately to understand whether its churn rate differs from other age groups.

Visual 4 — Customer Churn Rate by Geography



Objective: To compare the percentage of customers who exited the bank across France, Germany and Spain.

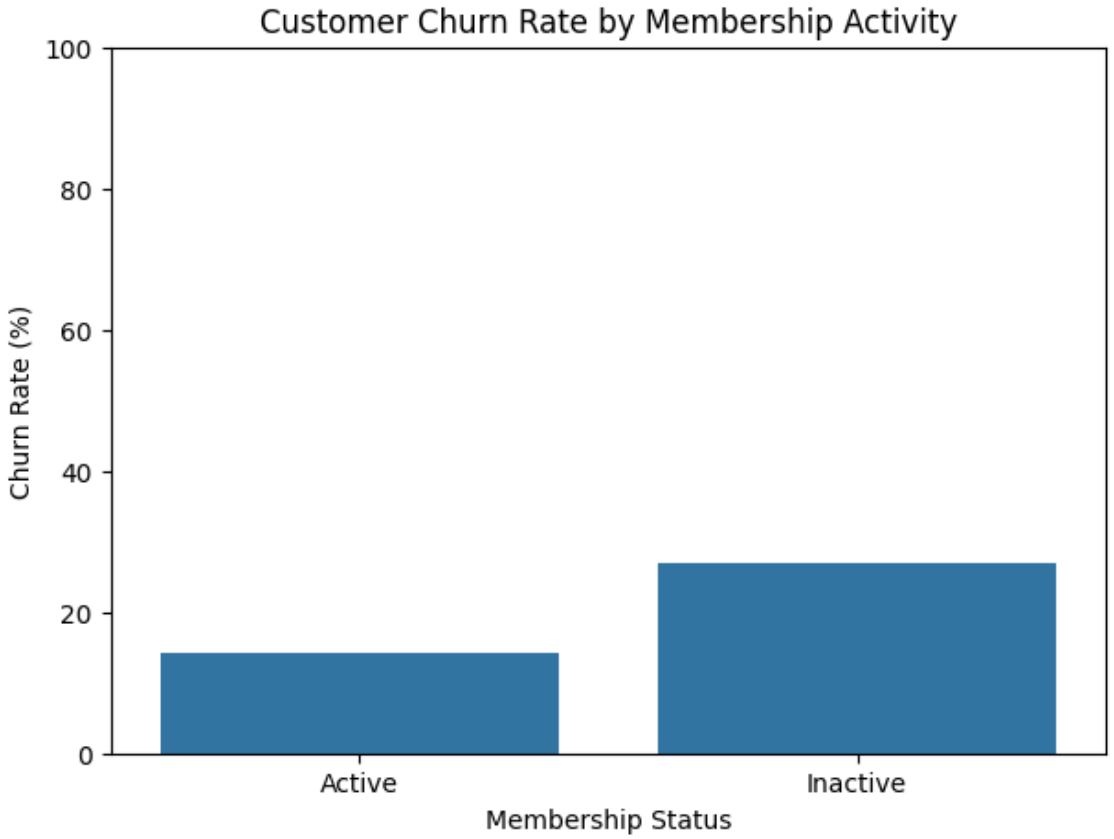
Visualization: Visual 4 — Customer Churn Rate by Geography (chart screenshot from the Python notebook).

Interpretation: Germany recorded the highest churn rate at 32.44%. Spain recorded 16.67%, while France recorded 16.15%. Germany's churn rate is approximately twice the rates observed in France and Spain.

Key Insight: Germany has the highest observed country-level churn rate, at 32.44%, compared with approximately 16% in France and Spain.

Business Insight: Prioritize further investigation of customer experience, product suitability, service expectations and local market conditions in Germany. The chart identifies where to investigate, not why customers leave.

Visual 5 — Customer Churn Rate by Membership Activity



Objective: To determine whether customer churn rates differ between active and inactive bank members.

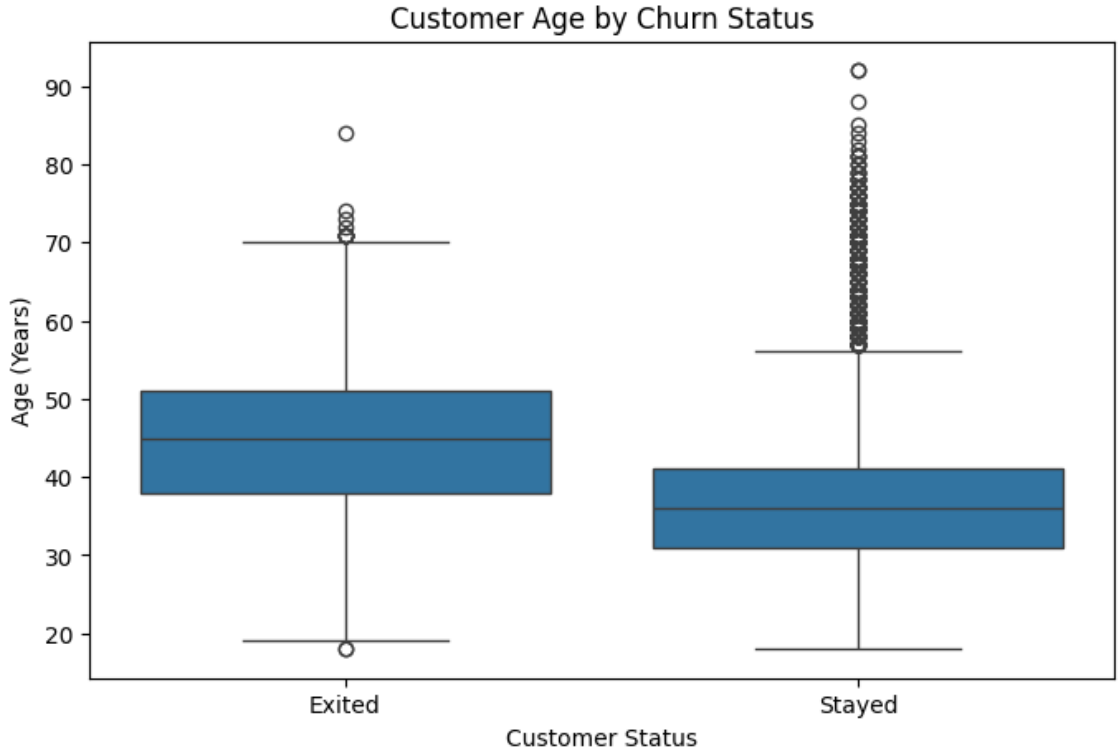
Visualization: Visual 5 — Customer Churn Rate by Membership Activity (chart screenshot from the Python notebook).

Interpretation: Inactive members had a churn rate of 26.85%, compared with 14.27% among active members. This is a difference of 12.58 percentage points; the inactive-member churn rate is approximately 1.88 times the active-member rate.

Key Insight: Inactive members show substantially higher churn than active members in this dataset.

Business Insight: Investigate why inactive members are less engaged and test targeted engagement or service initiatives. Measure subsequent churn to determine whether interventions are effective.

Visual 6 — Customer Age by Churn Status



Objective: To compare customer age distributions between customers who stayed and customers who exited.

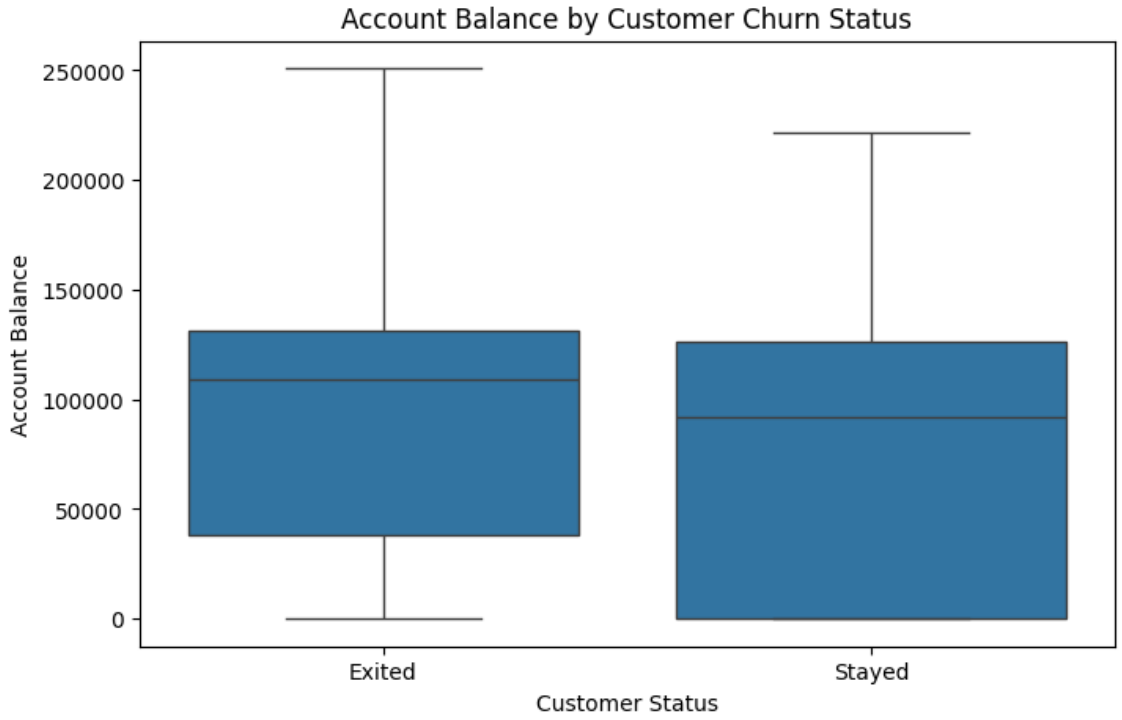
Visualization: Visual 6 — Customer Age by Churn Status (chart screenshot from the Python notebook).

Interpretation: Exited customers had an average age of 44.84 years and a median age of 45 years. Customers who stayed had an average age of 37.41 years and a median age of 36 years. The exited group was therefore older by 7.43 years on average.

Key Insight: Customers who exited were older on average than customers who stayed.

Business Insight: Review the needs and experiences of older customer segments, especially where higher churn is also observed. Age is an associated characteristic, not a demonstrated cause of churn.

Visual 7 — Account Balance by Customer Churn Status



Objective: To compare account-balance distributions between customers who stayed and customers who exited.

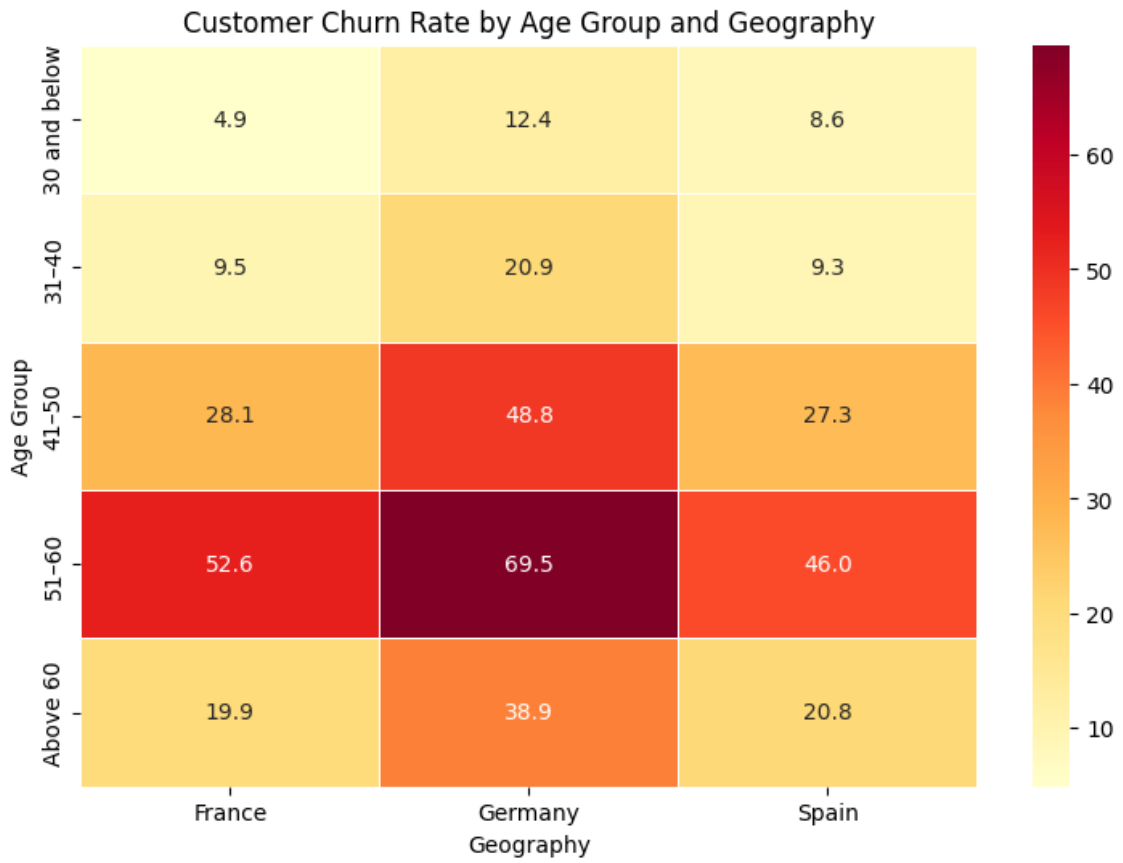
Visualization: Visual 7 — Account Balance by Customer Churn Status (chart screenshot from the Python notebook).

Interpretation: Exited customers had an average balance of 91,108.54 and a median balance of 109,349.29. Customers who stayed had an average balance of 72,745.30 and a median balance of 92,072.68. The exited group had higher average and median balances.

Key Insight: Customers who exited held higher average and median account balances than customers who stayed.

Business Insight: Investigate whether higher-balance customers have different product expectations, service needs or experiences. The observed difference does not mean high balances cause churn.

Visual 8 — Churn Rate by Age Group and Geography



Objective: To examine how churn rates vary across combinations of customer age groups and geographical regions.

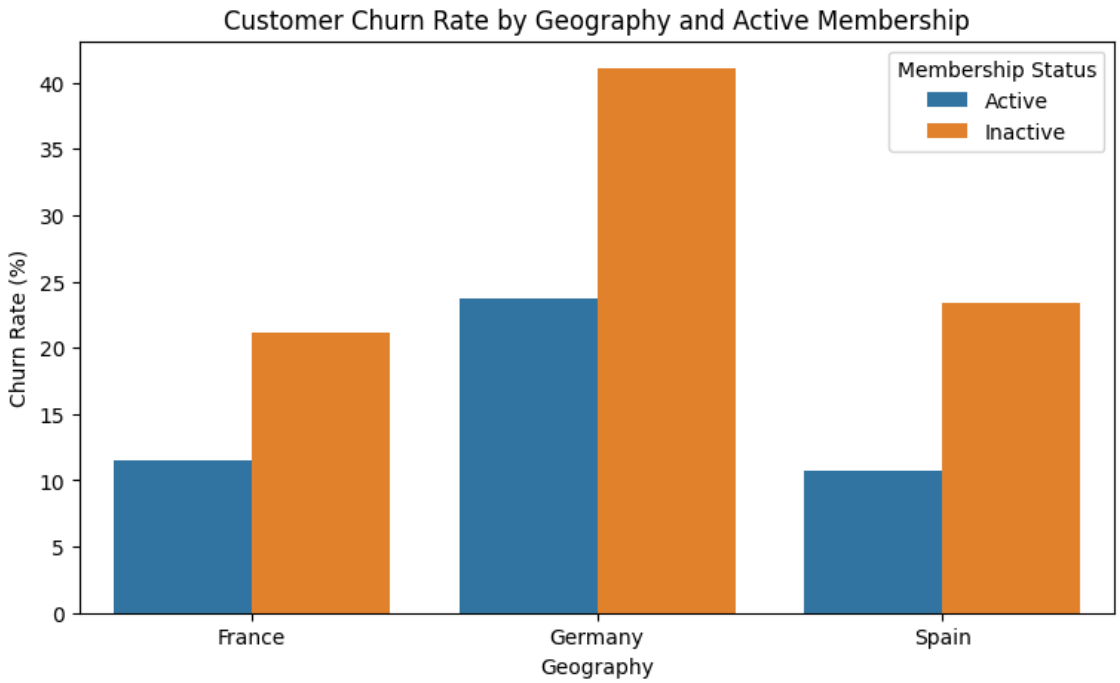
Visualization: Visual 8 — Churn Rate by Age Group and Geography (chart screenshot from the Python notebook).

Interpretation: The highest observed rate was 69.55% among customers aged 51–60 in Germany. Customers aged 51–60 in France had a churn rate of 52.59%, while customers aged 41–50 in Germany had a rate of 48.82%.

Key Insight: The 51–60 age group in Germany has the highest observed churn rate among the age-group and geography combinations shown.

Business Insight: Investigate this segment as a retention priority, but review the number of customers in each age-country group before allocating resources. A high rate in a small group may be less stable.

Visual 9 — Churn Rate by Geography and Active Membership



Objective: To compare churn rates across countries while separating active and inactive members.

Visualization: Visual 9 — Churn Rate by Geography and Active Membership (chart screenshot from the Python notebook).

Interpretation: Inactive members had higher churn in every country. Germany recorded 41.08% for inactive members and 23.72% for active members. Spain recorded 23.35% and 10.75%, respectively; France recorded 21.13% and 11.50%.

Key Insight: The higher churn rate among inactive members appears in all three countries, with Germany showing the highest rates for both membership groups.

Business Insight: Investigate engagement and retention practices across countries, with particular attention to inactive members in Germany. Test and monitor targeted interventions rather than assuming inactivity is the cause.

10. Overall Key Findings

- Overall, 2,037 of 10,000 customers exited (20.37%), while 7,963 stayed (79.63%).
- Germany had the highest overall churn rate at 32.44%, compared with 16.67% in Spain and 16.15% in France.
- Inactive members had a churn rate of 26.85%, compared with 14.27% among active members, a difference of 12.58 percentage points.
- Exited customers were older on average (44.84 years) than customers who stayed (37.41 years).
- Exited customers had a higher average account balance (91,108.54) than customers who stayed (72,745.30).
- The highest age-geography churn rate shown was 69.55% among customers aged 51–60 in Germany.
- Inactive members had higher churn than active members in all three countries; Germany had the highest rates for both groups.

11. Business Recommendations

1. Investigate retention challenges in Germany, where the overall churn rate is highest. Review customer experiences, product suitability, service expectations and local market conditions.
2. Investigate why inactive members have higher churn rates. Test appropriate engagement initiatives and measure whether they improve retention.
3. Examine the needs and experiences of customers aged 41–60, particularly in Germany. Check segment sizes before prioritizing actions.
4. Investigate the service needs and experiences of higher-balance customers who exited.
5. Monitor overall and segment-level churn rates over time and evaluate retention initiatives using subsequent customer data.

12. Limitations

- The dataset does not contain customers' stated reasons for leaving, so exact causes of churn cannot be identified.
- The analysis identifies associations and group differences, not causal relationships.
- The original data collection period is not specified in the supplied dataset information.
- Some age-geography combinations may contain fewer customers than others; group sizes should be reviewed before prioritizing actions.
- No predictive machine-learning model was built or evaluated in this project.

13. Future Enhancements

6. Add customer transactions, customer-service interactions, complaints and exit-survey information to investigate reasons for churn more directly.
7. Use a larger or more recent dataset and monitor churn over time.
8. Conduct statistical significance tests for selected group comparisons.
9. Develop interactive dashboards using Power BI or Tableau.
10. Build and evaluate classification models such as Logistic Regression, Decision Tree and Random Forest using precision, recall, F1-score, ROC-AUC and a confusion matrix.
11. Test customer-retention initiatives and measure their outcomes.

14. Conclusion

The Bank Customer Churn Analysis project demonstrates how Python, statistical analysis and visualization can be used to investigate customer attrition patterns. The dataset contains 10,000 customer records and has an overall churn rate of 20.37%. The analysis identifies higher churn in Germany (32.44%), among inactive members (26.85%), and among selected age groups. Customers who exited were generally older and had higher average and median account balances than customers who stayed.

The combined age-group and geography analysis highlighted customers aged 51–60 in Germany, with an observed churn rate of 69.55%, as a segment warranting further investigation. Inactive members in Germany also showed a high churn rate of 41.08%. These findings help identify where the bank may focus further investigation into customer experiences and retention opportunities.

The dataset does not record why customers left, so the findings should be interpreted as associations rather than proven causes. Overall, the project demonstrates practical skills in data inspection, cleaning, transformation, exploratory data analysis, statistical interpretation, outlier detection, visualization and business-focused reporting.

Future work could incorporate customer feedback and transaction data, test retention initiatives, and develop predictive churn models.